MA1522 Linear Algebra for Computing

Lesson 8: Linear Spans and Subspaces

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson8` (e.g. `A1234567Z_lesson8.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 8**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 3.3 Comparing Two Spans

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $T = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be sets of vectors in $\mathbb{R}^n$.

Theorem

$$\text{span}(T) \subseteq \text{span}(S) \iff \left(\begin{array}{c|c} \mathbf{u}_1 & \cdots \; \mathbf{u}_k \end{array} \right. \left| \begin{array}{c} \mathbf{v}_1 \end{array} \right. \left. \begin{array}{c} \cdots \; \mathbf{v}_m \end{array} \right) \text{ is consistent.}$$

So to check whether $\text{span}(S) = \text{span}(T)$, reduce twice:

$$\left(\begin{array}{c|c} S & T \end{array} \right) \text{ consistent} \iff \text{span}(T) \subseteq \text{span}(S),$$

$$\left(\begin{array}{c|c} T & S \end{array} \right) \text{ consistent} \iff \text{span}(S) \subseteq \text{span}(T).$$

Question 1

Alice and Bob each solve the same homogeneous system and write the general solution as

Alice: $s\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}-1\\0\\1\end{pmatrix}, \quad s,t\in\mathbb{R},$

Bob: $p\begin{pmatrix}1\\2\\1\end{pmatrix}+q\begin{pmatrix}1\\3\\2\end{pmatrix}, \quad p,q\in\mathbb{R}.$

Do Alice and Bob describe the same set of solutions?

Question 1 — solution

Each answer is a linear span: Alice's set is $\mathrm{span}(S)$ and Bob's is $\mathrm{span}(T)$, where $S = \{(1,1,0)^T, (-1,0,1)^T\}$ and $T = \{(1,2,1)^T, (1,3,2)^T\}$. So the question is whether $\mathrm{span}(S) = \mathrm{span}(T)$, and that is the algorithm of the recall slide: reduce twice.

$$(S \mid T) = \left(\begin{array}{cc|cc} 1 & -1 & 1 & 1 \\ 1 & 0 & 2 & 3 \\ 0 & 1 & 1 & 2 \end{array}\right) \xrightarrow[R_1+R_2]{R_2-R_1,\ R_3-R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

$$(T \mid S) = \left(\begin{array}{cc|cc} 1 & 1 & 1 & -1 \\ 2 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow[R_3-R_2,\ R_1-R_2]{R_2-2R_1,\ R_3-R_1} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Both are consistent, so $\mathrm{span}(T) \subseteq \mathrm{span}(S)$ and $\mathrm{span}(S) \subseteq \mathrm{span}(T)$: **Alice and Bob agree.** The first reduction says how: $(1,2,1)^T = 2(1,1,0)^T + (-1,0,1)^T$ ✓ and $(1,3,2)^T = 3(1,1,0)^T + 2(-1,0,1)^T$.

□

Section 3.4 Subspaces

Definition

A subset V of $\mathbb{R}^n$ is a **subspace** if

(i) $\mathbf{0} \in V$;

(ii) V is closed under scalar multiplication: $\alpha\mathbf{v} \in V$ for all $\mathbf{v} \in V$ and $\alpha \in \mathbb{R}$;

(iii) V is closed under addition: $\mathbf{u} + \mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$.

Equivalently, V is a subspace if

(i′) V is nonempty;

(ii′) V is closed under linear combinations: $\alpha\mathbf{u} + \beta\mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$ and all $\alpha, \beta \in \mathbb{R}$.

To show that V is not a subspace, exhibit one of the following: $\mathbf{0} \notin V$; a vector $\mathbf{v} \in V$ and a scalar α with $\alpha\mathbf{v} \notin V$; or vectors $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{u} + \mathbf{v} \notin V$.

Sections 3.3 & 3.4 Properties of Linear Spans

Theorem

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a set of vectors in $\mathbb{R}^n$.

(i) The zero vector $\mathbf{0}$ is in span(S).

(ii) span(S) is closed under scalar multiplication: if $\mathbf{u} \in$ span(S) and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{u} \in$ span(S).

(iii) span(S) is closed under addition: if $\mathbf{u}, \mathbf{v} \in$ span(S), then $\mathbf{u} + \mathbf{v} \in$ span(S).

Namely, span(S) is a subspace.

Theorem (Section 3.4: equivalent definition for subspaces)

A subset $V \subseteq \mathbb{R}^n$ is a subspace if and only if it is a linear span, $V =$ span(S), for some finite set $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ of vectors in $\mathbb{R}^n$.

Question 2

Determine whether each of the following subsets of $\mathbb{R}^3$ is a subspace.

1. $V_1 = \left\{ (x, y, z)^T \in \mathbb{R}^3 \mid x = 2y \right\}$
2. $V_2 = \left\{ (x, y, z)^T \in \mathbb{R}^3 \mid x \geq 0 \right\}$
3. $V_3 = \left\{ (x, y, z)^T \in \mathbb{R}^3 \mid xy = 0 \right\}$

If it is a subspace, express it as a linear span. If it is not, say which of the three conditions fails, and give an explicit vector or pair of vectors that shows it.

Question 2 — solution: V_1 and V_2

V_1 **is a subspace.** The condition $x = 2y$ says every vector of V_1 has the form

$$\begin{pmatrix} 2y \\ y \\ z \end{pmatrix} = y \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \qquad y, z \in \mathbb{R}, \qquad \text{so} \qquad V_1 = \operatorname{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\},$$

and every linear span is a subspace.

V_2 is **not** a subspace. It contains $\mathbf{0}$, and it is closed under addition, since $x_1 \geq 0$ and $x_2 \geq 0$ give $x_1 + x_2 \geq 0$. But it is not closed under scalar multiplication:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \in V_2, \qquad \text{while} \qquad (-1) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \notin V_2.$$

Question 2 — solution: V_3

V_3 is **not** a subspace, but the first two conditions do hold.

(i) $\mathbf{0} \in V_3$, since $0 \cdot 0 = 0$.

(ii) If $(x, y, z)^T \in V_3$ and $\alpha \in \mathbb{R}$, then $(\alpha x)(\alpha y) = \alpha^2(xy) = \alpha^2 \cdot 0 = 0$, so $\alpha(x, y, z)^T \in V_3$.

(iii) fails:

$$\begin{pmatrix}1\\0\\0\end{pmatrix} \in V_3 \quad \text{and} \quad \begin{pmatrix}0\\1\\0\end{pmatrix} \in V_3, \qquad \text{but} \qquad \begin{pmatrix}1\\0\\0\end{pmatrix} + \begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix} \notin V_3,$$

since $1 \cdot 1 = 1 \neq 0$.

So each of the three conditions is doing work: V_2 fails only (ii) and V_3 fails only (iii). □

Section 3.4 Affine Spaces

Let $A\mathbf{x} = \mathbf{b}$ be consistent, let $\mathbf{u}$ be a **particular solution**, $A\mathbf{u} = \mathbf{b}$, and let $V = \{\, \mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \mathbf{0} \,\}$ be the solution space of the associated homogeneous system.

Theorem (Affine Space)

The solution set of $A\mathbf{x} = \mathbf{b}$ is

$$W = \mathbf{u} + V := \{\, \mathbf{u} + \mathbf{v} \mid \mathbf{v} \in V \,\}.$$

In general an affine space $\mathbf{u} + V$ is not a subspace: it is one exactly when $\mathbf{u} \in V$. Here $\mathbf{u} \in V$ means $A\mathbf{u} = \mathbf{0}$, that is $\mathbf{b} = \mathbf{0}$, and then $W = V$. For $\mathbf{b} \neq \mathbf{0}$ we get $\mathbf{0} \notin W$, and W is V shifted away from the origin by $\mathbf{u}$: the two are parallel.

Example

Let $\mathbf{u}_1 + W_1$ and $\mathbf{u}_2 + W_2$ be two affine spaces inside $\mathbb{R}^n$. Here $\mathbf{u}_1, \mathbf{u}_2$ are two vectors of $\mathbb{R}^n$ and W_1, W_2 are subspaces of $\mathbb{R}^n$. True or false?

1. If $\mathbf{u}_1 + W_1 = \mathbf{u}_2 + W_2$ as subsets of $\mathbb{R}^n$, then $\mathbf{u}_1 = \mathbf{u}_2$.
2. If $\mathbf{u}_1 + W_1 = \mathbf{u}_2 + W_2$ as subsets of $\mathbb{R}^n$, then $W_1 = W_2$.

(1) **False.** In $\mathbb{R}^2$ take $W_1 = W_2 = \text{span}\{(1,0)^T\}$, $\mathbf{u}_1 = (0,0)^T$ and $\mathbf{u}_2 = (1,0)^T$: both sets are the x-axis, but $\mathbf{u}_1 \neq \mathbf{u}_2$.

(2) **True.** Let X be the common set and take $\mathbf{w}_1 \in W_1$. Both $\mathbf{u}_1 + \mathbf{w}_1$ and $\mathbf{u}_1$ lie in $X = \mathbf{u}_2 + W_2$, say $\mathbf{u}_1 + \mathbf{w}_1 = \mathbf{u}_2 + \mathbf{w}$ and $\mathbf{u}_1 = \mathbf{u}_2 + \mathbf{w}'$ with $\mathbf{w}, \mathbf{w}' \in W_2$. Subtracting, $\mathbf{w}_1 = \mathbf{w} - \mathbf{w}' \in W_2$, since W_2 is closed under linear combinations. So $W_1 \subseteq W_2$, and by symmetry $W_1 = W_2$.

An affine space determines its subspace, but not the vector it is shifted by.

Challenge

Let W be a subspace of $\mathbb{R}^n$ and let $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^n$. Show that

$$\mathbf{u}_1 + W = \mathbf{u}_2 + W \quad \Longleftrightarrow \quad \mathbf{u}_1 - \mathbf{u}_2 \in W.$$

($\Rightarrow$) Since $\mathbf{0} \in W$, we have $\mathbf{u}_1 = \mathbf{u}_1 + \mathbf{0} \in \mathbf{u}_1 + W = \mathbf{u}_2 + W$, so $\mathbf{u}_1 = \mathbf{u}_2 + \mathbf{w}$ for some $\mathbf{w} \in W$. Hence $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w} \in W$.

($\Leftarrow$) Suppose $\mathbf{u}_1 - \mathbf{u}_2 \in W$. Any element of $\mathbf{u}_1 + W$ is $\mathbf{u}_1 + \mathbf{w}$ with $\mathbf{w} \in W$, and

$$\mathbf{u}_1 + \mathbf{w} = \mathbf{u}_2 + \big((\mathbf{u}_1 - \mathbf{u}_2) + \mathbf{w}\big) \in \mathbf{u}_2 + W,$$

since W is closed under addition. So $\mathbf{u}_1 + W \subseteq \mathbf{u}_2 + W$. Also $\mathbf{u}_2 - \mathbf{u}_1 = (-1)(\mathbf{u}_1 - \mathbf{u}_2) \in W$, because W is closed under scalar multiplication, so the same argument with the roles swapped gives $\mathbf{u}_2 + W \subseteq \mathbf{u}_1 + W$. $\square$

Question 3

Alice and Bob each solve the same system $A\mathbf{x} = \mathbf{b}$, with $\mathbf{b} \neq \mathbf{0}$, and write the general solution as

$$\text{Alice:} \quad \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \qquad \text{Bob:} \quad \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + p \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + q \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix},$$

with $s, t, p, q \in \mathbb{R}$.

Do Alice and Bob describe the same set of solutions?

Question 3 — solution

Write $\mathbf{u}_S = (3,0,0)^T$ and $\mathbf{u}_T = (0,-2,0)^T$. Two things to check, in this order.

Are the two solution spaces the same? The direction sets are S and T of Question 1, where $\operatorname{span}(S) = \operatorname{span}(T)$ was proved. ✓ So both answers are affine spaces on the same subspace, $\mathbf{u}_S + \operatorname{span}(S)$ and $\mathbf{u}_T + \operatorname{span}(S)$.

Is $\mathbf{u}_T - \mathbf{u}_S$ in that subspace? Only now does the Challenge apply, with $W = \operatorname{span}(S)$: the two sets are equal exactly when $\mathbf{u}_T - \mathbf{u}_S = (-3,-2,0)^T$ lies in $\operatorname{span}(S)$.

$$\left(\begin{array}{cc|c} 1 & -1 & -3 \\ 1 & 0 & -2 \\ 0 & 1 & 0 \end{array}\right) \xrightarrow[R_1+R_2]{R_2-R_1,\ R_3-R_2} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{array}\right).$$

The last row reads $0 = -1$, so $\mathbf{u}_T - \mathbf{u}_S \notin \operatorname{span}(S)$ and **Alice and Bob do not agree**, although the first check passed. □

Next lesson

- **Sections 3.4 & 3.5 Subspaces and Linear Independence**

Please watch the video clips for these sections beforehand.